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Task Description for M.Sc. thesis of Jeanne Paulie Yap:

The integration of robotic systems in agriculture presents a promising solution to several issues, including enhancing production efficiency and reducing human labor costs. These technologies offer an opportunity to automate processes like ripeness monitoring, phenotyping, and harvesting, significantly transforming traditional agricultural management. However, despite these potential benefits, several challenges remain. One of the key issues is the complexity of agricultural environments, such as crop fields or greenhouses. In these settings, the robots are often faced with occlusions from other crops, leaves, or environmental factors, making it difficult to acquire complete crop data.

This thesis focuses on addressing these challenges that robotic systems may face while collecting data. Due to occlusions and other visual obstructions in the field or greenhouse, it is often impossible for robotic sensors to capture the entire structure of a plant or crop. This limitation impacts the system's ability to provide detailed information on crops, which is essential for effective crop management.

This thesis will investigate the task of estimating both 3D structure and the RGB colors of crops based on incomplete data. By predicting the shape and color of a plant from partial observations, it is possible to provide more detailed crop information without the need for full point cloud data, offering a more efficient and cost-effective solution for agricultural management. The thesis will investigate the use of a Conditional Generative Network for shape completion, leveraging the Denoising Diffusion Probabilistic Model (DDPM). This network can generate new samples from a target data used during training. This thesis will investigate the applicability of such models on generating a 3D shape of a crop based on partial shape observations, effectively "filling in" the missing parts. By conditioning the model on incomplete data, the system would infer the missing structure and RGB color information, overcoming the limitations posed by occlusions in traditional crop data collection methods.

The thesis will focus on 3D point clouds and partial-view RGB images of crops, s tarting with sweet pepper plants. Building upon the approach proposed by Nunes et al. [3], who applied DDPM for scene completion using LiDAR data, this thesis will extend their framework to agricultural applications. The study will adapt the DDPM model to predict both the complete 3D shape and the RGB colors of crops from partial observations. To achieve these results, the thesis will involve aligning the 3D point cloud with partial RGB

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data and training the model on this combined dataset, to achieve shape completion of crops.

Point Of Departure

Before beginning, it is essential to conduct a thorough literature review and become familiar with the data that will be used. Magistri et al. [1] proposed a dataset for shape completion for sweet pepper data. This dataset will be the starting point, later extending the method to different fruits. The next step will involve building upon the work of Nunes et al. [3], starting with the task of predicting the 3D shape of sweet pepper point clouds using the Denoising Diffusion Probabilistic Model (DDPM). Currently, the network operates solely on 3D point cloud data. The subsequent goal is to extend this by incorporating color information. This will be achieved by assigning RGB values to the point cloud based on paired point clouds and RGB images. This process will result in a new training set that combines both 3D and RGB information, enabling more comprehensive crop data analysis. Finally, the method will be evaluated using the sweet pepper dataset [1] and compared against state-of-the-art methods [2][4]. Also, the model will be trained on further data from other varieties of fruits, to evaluate the proposed shape completion approach on different fruits.

Kind regards,

Cyrill Stachniss

[1] F. Magistri, T. Läbe, E. Marks, S. Nagulavancha, Y. Pan, C. Smitt, L. Klingbeil, M. Halstead, H. Kuhlmann, C. McCool, J. Behley, and C. Stachniss. A Dataset and Benchmark for Shape Completion of Fruits for Agricultural Robotics. *arXiv preprint*, 2024.

[2] F. Magistri, E. Marks, S. Nagulavancha, I. Vizzo, T. Läbe, J. Behley, M. Halstead, C. McCool, and C. Stachniss. Contrastive 3D Shape Completion and Reconstruction for Agricultural Robots using RGB-D Frames. *IEEE Robotics and Automation Letters (RA-L)*, 7(4):10120–10127, 2022.

[3] L. Nunes, R. Marcuzzi, B. Mersch, J. Behley, and C. Stachniss. Scaling Diffusion Models to Real-World 3D LiDAR Scene Completion. In *Proc. of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[4] Y. Pan, F. Magistri, T. Läbe, E. Marks, C. Smitt, C. McCool, J. Behley, and C. Stachniss. Panoptic Mapping with Fruit Completion and Pose Estimation for Horticultural Robots. In *Proc. of the IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems (IROS)*, 2023.